

Suction type de-soldering stations

Suction type de-soldering stations are used primarily to de-solder through-hole components or to remove solder from the pads after the SMD component is removed with the help of SMD de-soldering equipment. These de-soldering stations are based on the principle of suction of the melted solder from the PCB with the help of vacuum. Its heated nozzle is put over a joint with the pin at the centre to melt the solder and, once it has melted, a button is pressed to initiate the suction of the melted solder.

There is a huge variety of such de-soldering stations available in the market, but they mainly fall into two categories:

1. Integrated de-soldering units
2. Vacuum pump type de-soldering stations

Integrated de-soldering units. These are equipped with a nozzle, built-in heating element and suction unit, and most of them run directly from the mains. Such units are comparatively cheaper than a complete station but lack various features. Most of them do not even have the temperature-control function. So the tip can go up to very high temperatures while in the stand. And when you try to de-



Fig. 2: Integrated type de-soldering unit

solder the joint you might even take out the pad—permanently damaging the board. Fig. 2 shows an integrated type de-soldering unit.

Vacuum-pump type de-soldering stations. These also work on the same principle as that of integrated units, but the power and suction is delivered

TABLE I
Solder Alloys and Their Melting Temperature

Solder alloy	Melting point (°C)
55Sn-95Pb	307
0.5Sn-92.5Pb-2.5Ag	280
95Sn/5Sb	243
100Sn/3	232
99.3Sn-0.7Cu	227
96.5Sn-3.5Ag	221
96.5Sn/3.0Ag/0.5Cu	219
95.2Sn/3.8Ag/1.0Cu	217
92.5Sn/3.5Ag/1.0Cu/3Bi	213
50In-50Pb	209
45Sn-55Pb	204
55Sn-45Pb	193
60Sn-40Pb	186
63Sn-37Pb	183
62Sn-36Pb-2Ag	179
97In-3Ag	143
43Sn/57Bi	139
52In-48Sn	118

through a different unit and is not integrated in the hand piece. Such stations come with better features like a digital display to ensure easy and reliable temperature control, power saving, auto shut-off and excellent thermal recovery. Such a unit can make your rework efficient and safe. Fig. 3 shows a vacuum pump type de-soldering station.

If you choose to buy a vacuum pump type de-soldering station, the next thing you need to decide on is the type of hand piece. There are different types of hand pieces available, such as gun type, pen type, straight, etc, and



Fig. 3: Vacuum pump type de-soldering station

everyone will have a different opinion on which is the best. So go by your own gut feeling. Personally, I find the pen type more comfortable for precision work.

Once you have made all the above decisions, it's time to work out the specifications for each model and choose one that best suits your requirements. Specifications for a de-soldering station are not too complicated but understanding them can make a big difference in making the right selection. Given below are the specifications and their descriptions to help you understand what they indicate about the de-soldering station.

Temperature control. The simplest and the cheapest de-soldering stations do not have any kind of thermal regulation. They are directly plugged into the wall socket and heat up to high temperatures in their holders. There is some amount of natural thermal regulation due to the physical environment. As the tip gets hotter, it dissipates more heat, and as it gets cooler, it dissipates less heat. Such stations may be described as thermally balanced but their output is neither controlled nor stable. Their high temperatures can cause permanent damage to the PCB.

Power consumption. When the nozzle tip touches the joint, its temperature drops, as the heat is dissipated in the joint. If the part you are soldering can dissipate heat at a higher rate than what your tip can deliver, the temperature will keep dropping until it won't melt solder any more. The wattage that you would require completely depends on the project at hand. Usually, bigger components absorb and dissipate more heat. If this rate is more than the heating rate of the de-soldering equipment, the joint starts cooling down. So the general rule is that you need more wattage for larger parts.

A higher wattage does not mean that the nozzle tip runs hotter. Instead, the heating is faster and there is more power in reserve, when required. Choose a de-soldering station with sufficient wattage to cover all